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A convex-relaxation based method for optimal water-power flow

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Abstract

This paper proposes a convex reformulation for the non-linear optimal water-power flow (OWPF) problem to optimize the operation cost of the integrated electricity–water network (IEWN). Due to the significant challenge in seeking the optimal solution of non-convex problems, the original OWPF problem is reformulated as a tractable mixed-integer second-order cone programming (MISOCP) problem. For the power distribution network, a second-order cone (SOC) relaxation is employed to address the non-linear branch power flow equation. For the water distribution network, a big-M trick is introduced to handle the unknown direction of water flow, then SOC relaxations and convex envelopes are employed to deal with quadratic and bilinear terms in non-convex constraints, respectively. To enhance the approximation accuracy of the proposed MISOCP model, a convex combination method and a sequential convexification approach are developed. Numerical results demonstrate that the proposed method outperforms the original non-linear formulation and linearized reformulation in accuracy, efficiency, and robustness.

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Keywords: Water network; Optimal water-power flow; SOC relaxation; Convex envelope; Sequential convexification

1. Introduction

In the context of the integrated energy system (IES), there is a growing interest in the electricity–water interdependences, the coordinated management and operation of the power system and water network recently. The optimal water-power flow (OWPF) problem is one of the fundamental coordinated operation issues for an integrated electricity–water network (IEWN). However, the OWPF problem is generally challenging to solve due to the non-convexities in non-linear alternating current (AC) power flow equations and water flow models. Researchers have devoted their efforts to the modeling and algorithm for solving OWPF problems. Oikonomou et al. [1] proposed

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Nomenclature

Sets

E_P	Edge set of power network
E_W	Edge set of water network
V_P	Bus set of power network
V_W	Node set of water network
W_L	Set of water pipes
W_P	Set of water pumps

Parameters

$C_{i,G}$	Cost of power generation output
$C_{m,R}$	Cost of water supplied by reservoirs
g	Gravity
h_m	Elevation of node m
$q_{m,L}$	Water load at node m
r	Formula coefficient (Darcy–Weisbach)
r_{ij}	Resistance of power line (i, j)
x_{ij}	Reactance of power line (i, j)
η	Pump efficiency
ρ	Water density

Variables

I_{mn}^+ / I_{mn}^-	Binary variables describing the direction of water flow
P_{ij} / Q_{ij}	Active/Reactive power flow of power line (i, j)
$P_{j,G} / Q_{j,G}$	Active/Reactive power generation output at bus j
$P_{mn,p}$	Power consumed by the pump (m, n)
$h_{mn,p}$	Head-gain supplied by the pump (m, n)
l_{ij}	Squared current magnitude of power line (i, j)
p_m	Pressure at node m
$q_{m,R}$	Water supplied at node m
q_{mn}	Volumetric water flow rate through pipe (m, n)
$q_{mn,p}$	Water flow rate entering the pump (m, n)
u_i	Squared voltage magnitude v_i^2 at bus i
$\omega_{mn,p}$	Relative speed of the pump (m, n)

a multi-period OWPF model for optimizing the energy flexibility of the water distribution system, in which the direct current (DC) power flow equation is adopted to describe the power system and the piecewise linearization is employed to deal with the non-linear hydraulic equations and variable-speed pump functions. Wang et al. [2] proposed a robust OWPF model based on the DC power flow equation for multiple energy systems, in which a linear model is used to model the pump head-gain function for simplification. These studies on the OWPF problem mainly focus on the coupling of the transmission power system and water network because of the DC power flow equation used, whereas the interdependence of the power distribution network and water distribution network for urban areas is usually stronger, indicating the necessity for OWPF analysis of the distribution-level IEWN.

As for distribution-level OWPF studies, Zamzam et al. [3] introduced a feasible point pursuit-successive convex approximation for the non-convex OWPF problem with the assumption of fixed water flow directions. However, the water flow directions are not always known but can be adjusted for the operational requirement of the water network. Li et al. [4] proposed a quasi-convex hull-based OWPF model, where a simplified model is adopted for the pump

head-gain with the assumption of fixed-speed pumps (FSP) that can reduce the modeling complexity. Therefore, addressing the non-linear model of variable-speed pumps (VSP) is necessary. Less work considered unknown water flow directions and VSPs at the same time, Sui et al. [5] proposed an electric-hydraulic dispatching model for a smart water-energy community, where piecewise linearization was adopted for the non-linear pressure head-loss and variable speed pump equations. The work of [1] also introduced piecewise linearization to deal with the non-linearity. However, a higher accuracy will require numerous piecewise segments and introduce a large number of binary variables, which may limit the scalability of the piecewise linearization method.

As discussed above, the distribution-level OWPF problem has not been thoroughly studied, especially for the treatments on unknown water flow directions and the model of VSPs. They are either unilaterally studied or both handled by piecewise linearization, which is ineffective when high-accuracy solutions are required.

Therefore, this paper studies the distribution-level OWPF problem considering the variable-speed pumps and unspecified water flow directions, and *proposes a convex relaxation-based method for solving the OWPF problem considering VSPs and unknown water flow directions*. The OWPF problem is reformulated as a tractable mixed-integer SOC programming (MISOCP) problem. A sequential convexification approach and a convex combination method are also developed to enhance the tightness of SOC relaxations used and the approximation performance of the convex envelope, respectively. The proposed convex relaxation-based OWPF method has superior computation performance in accuracy, efficiency, and scalability which is demonstrated by simulations and analysis and compared with the non-linear formulation and the linearized reformulation used in [1].

2. The non-linear OWPF problem

2.1. Power distribution network constraints

The DistFlow model [6] is adopted here to model the power distribution network:

$$P_{ij} = \sum_{k: j \rightarrow k} P_{jk} + r_{ij} l_{ij} + P_{j,L} - P_{j,G}, (i, j) \in E_P \quad (1)$$

$$Q_{ij} = \sum_{k: j \rightarrow k} Q_{jk} + x_{ij} l_{ij} + Q_{j,L} - Q_{j,G}, (i, j) \in E_P \quad (2)$$

$$u_i - u_j = 2(r_{ij} P_{ij} + x_{ij} Q_{ij}) - (r_{ij}^2 + x_{ij}^2) l_{ij}, (i, j) \in E_P \quad (3)$$

$$P_{ij}^2 + Q_{ij}^2 = l_{ij} u_i \quad (4)$$

where power equations are described by squared bus voltage magnitude u_i and branch power flow variables. Eqs. (1)–(2) determine the active/reactive power flowing through a power line, respectively. Eq. (3) describes the voltage drop along a power line. Eq. (4) describes the relationship of power flow, current magnitude and voltage magnitude. The constraints for power, and voltage magnitude are omitted due to limited space. It is noted that in this paper, overlines and underlines denote upper/lower bounds, respectively.

2.2. Water network constraints

The operation constraints of water distribution network are given as follows:

$$q_{m,R} = \sum_{(m,n) \in E_W} q_{mn} + q_{m,L} \quad (5)$$

$$(h_m + p_m) - (h_n + p_n) = \text{sgn}(q_{mn}) r q_{mn}^2, (m, n) \in W_L \quad (6)$$

$$-\bar{q}_{mn} \leq q_{mn} \leq \bar{q}_{mn}, (m, n) \in W_L \quad (7)$$

$$\underline{p}_m \leq p_m \leq \bar{p}_m, m \in V_W \quad (8)$$

$$h_{mn,p} = a q_{mn,p}^2 + b q_{mn,p} \omega_{mn,p} + c \omega_{mn,p}^2, (m, n) \in W_P \quad (9)$$

$$P_{mn,p} = \frac{\rho g}{\eta} q_{mn,p} h_{mn,p}, (m, n) \in W_P \quad (10)$$

where (5) describes the water flow balance at node m . If water flows from node m to n , q_{mn} is positive. Eq. (6) is the widely-used Darcy–Weisbach equation to calculate the head-loss along a pipe, where the coefficient r is determined by friction factor, pipe length, pipe diameter, etc [3]. Eqs. (7) and (8) represent the allowable pipe flow and node head, respectively. The electric motor-driven pump is the coupling element between the power system and the water system. The head-gain and power of the VSP is computed by (9) and (10), respectively [7].

2.3. Non-linear OWPF formulation

Based on Sections 2.1 and 2.2, the OWPF problem is formulated as

$$\begin{aligned} \min \quad & \sum_i C_{i,G} P_{i,G} + \sum_m C_{m,R} q_{m,R} \\ \text{s.t.} \quad & (1)–(10). \end{aligned} \quad (11)$$

The objective of the OWPF (11) is to minimize the overall cost of purchasing electricity and water. The non-convexities in (4), pipe head-loss function (6) and pump Eqs. (9)–(10) pose a significant challenge in seeking the optimal solution. The treatments for the non-convexity will be introduced in the coming section.

3. Convex reformulation for the non-convex OWPF model

3.1. Convex reformulation for the non-linear OWPF

(1) SOC relaxation of power flow equations

As for the non-convex equality constraint (4), a SOC relaxation is introduced to convert (4) into (12) [8].

$$P_{ij}^2 + Q_{ij}^2 \leq l_{ij} u_i \Rightarrow 4P_{ij}^2 + 4Q_{ij}^2 + (l_{ij} - u_i)^2 \leq (l_{ij} + u_i)^2 \quad (12)$$

Note that the SOC relaxation in (12) is exact under a strict condition that the power flow of the radial power distribution network cannot be reversed [9].

(2) SOC relaxation of pipe head-loss function

There is a sign function in the pipe head-loss function (6) that is intractable when solving the OWPF model, so we introduce two binary variables, I_{mn}^+ and I_{mn}^- to describe the direction of the water flow through the pipe (m, n) . If $q_{mn} > 0$, I_{mn}^+ equals 1; else if $q_{mn} < 0$, I_{mn}^- equals 1. The pipe head-loss function can be transformed as:

$$I_{mn}^+ + I_{mn}^- = 1, I_{mn}^+, I_{mn}^- \in \{0, 1\}, (m, n) \in W_L \quad (13)$$

$$-\bar{q}_{mn}(1 - I_{mn}^+) \leq q_{mn} \leq \bar{q}_{mn}(1 - I_{mn}^-), (m, n) \in W_L \quad (14)$$

$$r q_{mn}^2 = (I_{mn}^+ - I_{mn}^-)(H_m - H_n), (m, n) \in W_L \quad (15)$$

The constraint (15) still holds a complex form, which can be relaxed to a SOC relaxation (16). In (16), the introduced auxiliary variable γ_{mn} represent the right-hand side of (15) and is bounded by (17)–(18) using the big M trick and the McCormick envelop method [10]. Noted that M is a large number.

$$r q_{mn}^2 \leq \gamma_{mn}, (m, n) \in W_L \quad (16)$$

$$\begin{aligned} H_n - H_m - M(I_{mn}^+ - I_{mn}^- + 1) &\leq \gamma_{mn} \\ &\leq H_n - H_m + M(I_{mn}^+ - I_{mn}^- + 1), (m, n) \in W_L \end{aligned} \quad (17)$$

$$\begin{aligned} H_m - H_n + M(I_{mn}^+ - I_{mn}^- - 1) &\leq \gamma_{mn} \\ &\leq H_m - H_n - M(I_{mn}^+ - I_{mn}^- - 1), (m, n) \in W_L \end{aligned} \quad (18)$$

(3) Convex approximation for pump equations

The non-convexity in the pump function (9) is caused by the quadratic terms and the bilinear terms. We use auxiliary variables $\alpha_{mn,p}$ and $\chi_{mn,p}$ to substitute the quadratic terms $q_{mn,p}^2$, and $\omega_{mn,p}^2$, and $\beta_{mn,p}$ to substitute the bilinear terms $q_{mn,p}\omega_{mn,p}$, then relax each quadratic equality constraint with a SOC inequality and convex hull, as shown in (20)–(21). As for the bilinear term $q_{mn,p}\omega_{mn,p}$, the McCormick envelope formed by (22)–(23) is used to approximate it.

$$h_{mn,p} = a\alpha_{mn,p} + b\beta_{mn,p} + c\chi_{mn,p}, (m, n) \in W_P \quad (19)$$

$$q_{mn,p}^2 \leq \alpha_{mn,p} \quad (20)$$

$$\omega_{mn,p}^2 \leq \chi_{mn,p} \quad (21)$$

$$\begin{aligned} q_{mn,p} \cdot \underline{\omega}_{mn,p} + \underline{q}_{mn,p} \cdot \omega_{mn,p} - \underline{q}_{mn,p} \cdot \underline{\omega}_{mn,p} &\leq \beta_{mn,p} \\ &\leq q_{mn,p} \cdot \underline{\omega}_{mn,p} + \bar{q}_{mn,p} \cdot \omega_{mn,p} - \bar{q}_{mn,p} \cdot \underline{\omega}_{mn,p} \end{aligned} \quad (22)$$

$$\begin{aligned} q_{mn,p} \cdot \bar{\omega}_{mn,p} + \bar{q}_{mn,p} \cdot \omega_{mn,p} - \bar{q}_{mn,p} \cdot \bar{\omega}_{mn,p} &\leq \beta_{mn,p} \\ &\leq q_{mn,p} \cdot \bar{\omega}_{mn,p} + \underline{q}_{mn,p} \cdot \omega_{mn,p} - \underline{q}_{mn,p} \cdot \bar{\omega}_{mn,p} \end{aligned} \quad (23)$$

(4) Convex envelope for pump power equations

Likewise, the bilinear term in the pump power equation (10) can also be substituted by a new variable $\varphi_{mn,p}$, and relaxed by the McCormick Envelopes. Then, the pump power equation can be approximated by (24)–(26):

$$P_{mn,p} = \frac{\rho g}{\eta} \varphi_{mn,p}, (m, n) \in W_P \quad (24)$$

$$\begin{aligned} q_{mn,p} \cdot \underline{h}_{mn,p} + \underline{q}_{mn,p} \cdot h_{mn,p} - \underline{q}_{mn,p} \cdot \underline{h}_{mn,p} &\leq \varphi_{mn,p} \\ &\leq q_{mn,p} \cdot \underline{h}_{mn,p} + \bar{q}_{mn,p} \cdot h_{mn,p} - \bar{q}_{mn,p} \cdot \underline{h}_{mn,p} \end{aligned} \quad (25)$$

$$\begin{aligned} q_{mn,p} \cdot \bar{h}_{mn,p} + \bar{q}_{mn,p} \cdot h_{mn,p} - \bar{q}_{mn,p} \cdot \bar{h}_{mn,p} &\leq \varphi_{mn,p} \\ &\leq q_{mn,p} \cdot \bar{h}_{mn,p} + \underline{q}_{mn,p} \cdot h_{mn,p} - \underline{q}_{mn,p} \cdot \bar{h}_{mn,p} \end{aligned} \quad (26)$$

Based on the contents above, the non-linear OWPF (11) is reformulated as a MISOCP problem, referred as MISOCP-I in the following text.

3.2. The proposed enhanced convex reformulation

The OWPF problem MISOCP-I is tractable. However, the solution feasibility is related to the tightness of the employed McCormick envelope relaxation and the SOC relaxation. To tighten the relaxation and improve the solution feasibility, an enhanced convex reformulation for MISOCP-I, namely, MISOCP-II, is proposed herein.

(1) Enhanced McCormick envelope

For the McCormick envelope approximations of the bilinear terms $q_{mn,p}\omega_{mn,p}$ and $q_{mn,p}h_{mn,p}$, a convex combination approach in [11] is introduced here to improve the approximation performance.

Taking the bilinear term $q_{mn,p}\omega_{mn,p}$ as an illustration, we choose $q_{mn,p}$ to be the partitioned variable of which the domain is divided into U intervals. Each interval has an identical segment length of d , $d = (\bar{q}_{mn,p} - \underline{q}_{mn,p})/U$. So there are $U+1$ points that divide the domain with their values calculated by (27). A binary variable λ is introduced to model the partition of $q_{mn,p}$, $\lambda(u)=1$ if $q_{mn,p}$ locates at the u^{th} segment, i.e. $a(u) \leq q_{mn,p} \leq a(u+1)$. With global differential variables $\Delta q_{mn,p}$ and $\Delta \omega_{mn,p}$, we have $q_{mn,p}$ and $\omega_{mn,p}$ described by (29)–(30). Then $\Delta q_{mn,p}$ and $\Delta \omega_{mn,p}$ are bounded by (31)–(32). The McCormick Envelope constrain are replaced by (33)–(35) using $\Delta \beta_{mn,p}$, $\Delta q_{mn,p}$, and $\Delta \omega_{mn,p}$. Consequently, $\beta_{mn,p}$ can be expressed by a convex combination in (36).

$$a(u) = \underline{q}_{mn,p} + (u - 1) \cdot d, \forall u \leq U \quad (27)$$

$$\sum_{u=1}^U \lambda(u) = 1 \quad (28)$$

$$q_{mn,p} = \underline{q}_{mn,p} + d \cdot \sum_{u=1}^U [(u-1) \cdot \lambda(u)] + \Delta q_{mn,p} \quad (29)$$

$$\omega_{mn,p} = \underline{\omega}_{mn,p} + \sum_{u=1}^U \Delta \omega_{mn,p}(u) \quad (30)$$

$$0 \leq \Delta q_{mn,p} \leq d \quad (31)$$

$$0 \leq \Delta \omega_{mn,p}(u) \leq (\bar{\omega}_{mn,p} - \underline{\omega}_{mn,p}) \cdot \lambda(u) \quad (32)$$

$$\Delta \beta_{mn,p} \geq (\bar{\omega}_{mn,p} - \underline{\omega}_{mn,p}) \Delta q_{mn,p} + d(\omega_{mn,p} - \bar{\omega}_{mn,p}) \quad (33)$$

$$\Delta \beta_{mn,p} \leq d \cdot (\omega_{mn,p} - \underline{\omega}_{mn,p}) \quad (34)$$

$$\Delta \beta_{mn,p} \leq (\bar{\omega}_{mn,p} - \underline{\omega}_{mn,p}) \cdot \Delta q_{mn,p} \quad (35)$$

$$\begin{aligned} \beta_{mn,p} &= \underline{q}_{mn,p} \cdot \omega_{mn,p} + q_{mn,p} \cdot \underline{\omega}_{mn,p} - \underline{q}_{mn,p} \cdot \underline{\omega}_{mn,p} \\ &+ d \cdot \sum_{u=1}^U [(u-1) \cdot \Delta \omega_{mn,p}(u)] + \Delta \beta_{mn,p} \end{aligned} \quad (36)$$

(2) Convex-concave procedure (CCP) for tightening the SOC relaxation

The exactness of the SOC relaxations (12), (16), (20), and (21) in MISOCP-I is unpredictable and impacts the accuracy and feasibility of the solution. To tighten the relaxations, additional opposite constraints (37)–(40) are added.

$$4P_{ij}^2 + 4Q_{ij}^2 + (l_{ij} - u_i)^2 \geq (l_{ij} + u_i)^2 \quad (37)$$

$$rq_{mn}^2 \geq \gamma_{mn}, (m, n) \in W_L \quad (38)$$

$$q_{mn,p}^2 \geq \alpha_{mn,p} \quad (39)$$

$$\omega_{mn,p}^2 \geq \chi_{mn,p} \quad (40)$$

However, constraints (37)–(40) are concave. A penalty CCP [12] is introduced to address the concave constraints to ensure the tightness of the SOC relaxation. Concave constraints (37)–(40) hold the difference of convex form:

$$f(x) - g(x) \geq 0 \quad (41)$$

where $f(x)$ and $g(x)$ are convex.

The difference-of-convex form in (41) is convex when $f(x)$ is affine, but $f(x)$ in (37)–(40) are quadratic, which needs to be linearized. Given the linearized point $x^{(k)}$, the linear approximation for $f(x)$ is expressed as

$$f(x) \approx f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) \quad (42)$$

By combining (42) and introducing the additional positive slack variable s , (41) can be approximated by

$$g(x) - \left[f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) \right] \leq s \quad (43)$$

A sequential convexification method is then developed to address (43) to ensure the tightness of the SOC relaxation. The convergence proof can be found in [12].

Algorithm: A sequential convexification method

1. **Initialize.** Set the objective tolerance ε_o , maximal penalty coefficient $\tau_{\max}$, initial penalty coefficient $\tau_1 > 0$, updating coefficient $\mu > 1$, and iteration $k=1$. Solve MISOCP-I and set the solution $(q_{mn,p}^{(0)}, \omega_{mn,p}^{(0)}, q_{mn}^{(0)}, P_{ij}^{(0)}, Q_{ij}^{(0)}, I_{ij}^{(0)}, u_i^{(0)})$ as the linearization point of the first iteration.

2. Let $\mathbf{x}$ denote the vector of decision variables, and introduce a vector of slack variable $\mathbf{s}$ and a penalty coefficient τ to form the enhanced convex reformulation of the OWPF problem in a compact form.

$$\begin{aligned} \text{MISOCP-II: } \min_{\mathbf{x} \geq 0, \mathbf{s}} & \mathbf{C}^T \mathbf{x} + \tau \mathbf{1}^T \mathbf{s} \\ \text{s.t. } & \mathbf{Ax} \leq \mathbf{b}, \|\mathbf{Dx}\|_2 \leq \mathbf{Fs}, \|\mathbf{Gx}\|_2 \leq \mathbf{Jx} \end{aligned}$$

where coefficient vector/matrix $\mathbf{b}$, $\mathbf{A}$, $\mathbf{G}$, and $\mathbf{J}$ are obtained from OWPF constraints; $\mathbf{D}$ and $\mathbf{F}$ are coefficients for CCP constraints. Additionally, the second term in the objective function penalizes the increase of slack variables to tighten the quadratic constraints.

3. **While** $\tau \leq \tau_{\max}$ and $GAP_o \geq \varepsilon_o$, **Do**

4. Solve MISOCP-II and set the solution $(q_{mn,p}^{(k)}, \omega_{mn,p}^{(k)}, q_{mn}^{(k)}, P_{ij}^{(k)}, Q_{ij}^{(k)}, I_{ij}^{(k)}, u_i^{(k)})$ to the values of linearization point for the next iteration.

5. Update. $\tau_{k+1} = \min(\tau_{\max}, \mu \tau_k)$, $k = k+1$.

6. **End Do.**

Note that GAP_o in the algorithm is the relative error of objective value between two consecutive solutions.

Remark. The initial points for linearization in CCP may influence the performance of the algorithm. The solutions of MISOCP-I in Step 1 provide high-quality initial points.

4. Case study

Two case studies are used to verify the effectiveness of the proposed method. The first case study on a small-scale IEWN with the IEEE 13-node distribution network and an 11-node water network is implemented to analyze the tightness of the adopted relaxations, and test the solution accuracy and adaptability of the proposed method. The second one on a large-scale IEWN with the IEEE 123-bus power system and a 67-node water network [13] is used to demonstrate the scalability of the proposed method.

4.1. Small-scale IEWS test system

The detailed parameters of IEWN test system shown in Fig. 1 are available in [14].

(1) *Accuracy and efficiency analysis:* The following four methods are used for comparisons to illustrate the accuracy and efficiency of the proposed enhanced OWPF method.

M-I: Non-linear OWPF model introduced in Section 2, which is referred to as the benchmark.

M-II: Piecewise linearization based OWPF in [1]. The segment number of piecewise linearization for pipe head-loss, pump flow, and pump speed is respectively 20, 10, and 10.

M-III: The proposed convex reformulation of the OWPF in Section 3.1, i.e. MISOCP-I model.

M-IV: The proposed enhanced OWPF method in Section 3.2, i.e. the MISOCP-II model.

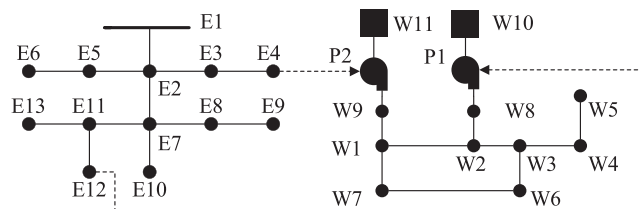


Fig. 1. The IEEE 13-bus power distribution network and 11-node water network.

We introduce two metrics to measure the accuracy of M-III and M-IV. One is the relative error on the objective function. Another is the maximum violation index of constraint to reflect the tightness of the relaxed quadratic

constraints. The violation indices of SOC constraints are calculated by (45)–(48).

$$\text{relative error} = |Obj - Obj_{\text{Benchmark}}| / Obj_{\text{Benchmark}} \times 100\% \quad (44)$$

$$V_{\text{power}} = \max \left\{ \left((l_{ij} + u_i)^2 - [4P_{ij}^2 + 4Q_{ij}^2 + (l_{ij} - u_i)^2] \right) / (l_{ij} + u_i)^2, (i, j) \in E_P \right\} \quad (45)$$

$$V_{\text{headloss}} = \max \{ (\gamma - r q_{mn}^2) / \gamma, (m, n) \in W_L \} \quad (46)$$

$$V_{\alpha} = \max \{ (\alpha_{mn,p} - q_{mn,p}^2) / \alpha_{mn,p}, (m, n) \in W \} \quad (47)$$

$$V_{\chi} = \max \{ (\chi_{mn,p} - \omega_{mn,p}^2) / \chi_{mn,p}, (m, n) \in W_L \} \quad (48)$$

As shown in Table 1, the results of M-IV are very close to M-I with very small relative errors, i.e. 0.005%, 0%, and 0.008% with respect to the total cost, water cost, and power cost. This indicates the accuracy of the proposed enhanced OWPF method (M-IV). Furthermore, the maximum violation indices of constraints that measure the tightness of the relaxations in M-IV are negligible (see Table 2), which demonstrates the exactness of the adopted relaxations with CCP and thus ensures the solution accuracy of M-IV. In comparison, the maximum violation indices of constraints in M-III are much larger than M-IV. The tremendous gaps degraded the tightness of the constraints, leading to an impractical solution. There exist evident relative errors in the optimized results compared to M-I (see Tables 1 and 2) since the relaxations in M-III without any tightening strategies are of poorer accuracy. Lower costs are found in M-III and M-IV because the SOC relaxations and convex envelopes expanded the feasible domain of decision-making compared with the original problem M-I, and better solutions may be found in these domains. As for M-II, the results of M-II in Table 1 are comparable to M-I with the root mean square error (RMSE) of pipe head-loss, pump head, and pump power respectively being 0.119, 0.007, and 0.013. The objective results and errors illustrate that the piecewise linearization in M-II can provide high-accuracy solutions.

Table 1. Cost results, gaps, and computational performance of the four methods under the small-scale IEWN.

Method	Total cost (\$)	Water cost (\$)	Power cost (\$)	Computation time (s)
M-I (Benchmark)	616.476	235.931	380.546	0.143
M-II	616.440	235.885	380.554	1.069
Relative error%	0.006	0.019	0.002	
M-III	607.931	230.699	377.232	0.247
Relative error%	1.386	2.217	0.871	
M-IV	616.446	235.931	380.516	0.516
Relative error%	0.005	0	0.008	

Table 2. Violations of constraints under the small-scale IEWN.

Method	V_{power}	V_{headloss}	V_{α}	V_{χ}
M-III	0.0350	1.0000	0.5006	0.2996
M-IV	4.00e−4	4.49e−06	4.87e−05	2.11e−07

In the computation efficiency aspect, among the four methods, M-I takes the least computational time according to Table 1. However, solving the non-linear model of M-I cannot guarantee the convergence of an optimal solution. Although M-II provides high-accuracy solutions, it takes the longest time 1.069 s to find a solution since the partition of non-linear functions introduces multiple integer variables. M-III is also fast, but it has been shown that the model lacks accuracy. M-IV takes a longer time 0.516s than M-III due to the iterative process. Taking both accuracy and efficiency into consideration, M-IV outperforms the other three methods, as it can always efficiently find an optimal solution with small errors.

(2) *Adaptability analysis:* This part aims to assess the adaptability of the proposed OWPF method in two cases.

Case 1: Heavy loading. Increasing both electricity and water loads of IEWN I to 1.7 times the original loads, we test the adaptability of the proposed method to high loading levels compared with the NLP model. Results and relative errors in total costs are shown in Table 3.

Table 3. Adaptability to heavy loads.

DemandMultiplier	Solvability		Iteration	Relative errors %
	M-I	M-IV	M-IV	
1.1	Feasible	Feasible	2	0.270
1.3	Feasible	Feasible	2	0.007
1.5	Feasible	Feasible	2	0.009
1.7	Infeasible	Feasible	2	–

The proposed method is comparable with M-I with small relative errors until loads are raised to 1.7 times the original load level. The noticeable relative errors of the scenario when the load is 1.1 times the initial load are due to a better solution the second-order cone programming (SOCP) model finds, where the pump operation (flow rate and speed) is optimized to achieve a lower water cost and power consumption. Besides, the violation indices are all kept under 10^{-4} . For the scenarios when IEWN I serves 1.7 times the original loads, the M-I fails to solve the problem, whereas the M-IV obtains an optimal solution with small gaps and primal residuals.

Case 2: Load fluctuation. In this case, we use the Monte Carlo method to generate 100 scenarios, where both electricity loads and water loads follow a normal distribution with the mean value being the initial load and the standard deviation being 10% of the mean value in this case. The solving status of the M-I and M-IV are displayed in Fig. 2, where a value 1 stands for finding an optimal solution successfully, and a value 0 stands for failure. The objective function gaps and violations are displayed in Figs. 3 and 4.

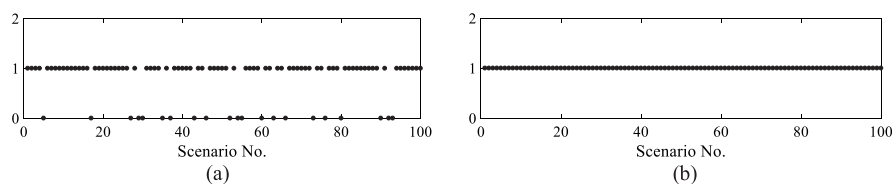


Fig. 2. The solvability of the different methods under load fluctuation: (a) the benchmark method; (b) the proposed method.

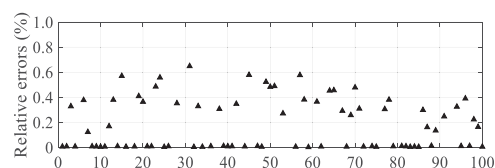


Fig. 3. Relative errors of the total cost.

Fig. 2 shows that the proposed method finds the feasible optimal solution in all simulations, whereas there are 21 scenarios when the solver IPOPT fails due to local infeasibility. The average iteration number of the proposed method is 2.15. Approximately half of the objective function gaps (i.e. the relative errors) of feasible solutions are above 0.1%, as shown in Fig. 3, while the violation indices of head-loss, pump head-gain, and pump power constraints are still small, as shown in Fig. 4. The results suggest that in these scenarios with large relative errors, M-IV obtains a better solution with all constraints tightened, which testified the excellent adaptability and relaxation tightness of the proposed method.

4.2. Large-scale IEWS test system

The scalability of the proposed M-III and M-IV is tested on a large-scale IEWN test system [14].

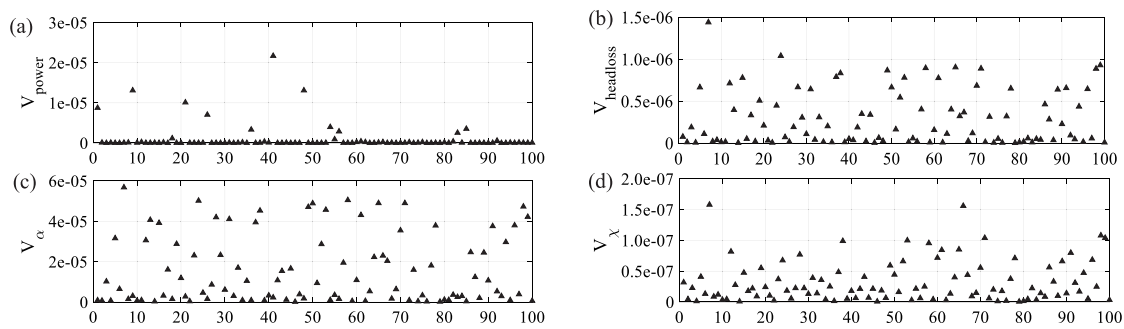


Fig. 4. Violations under load fluctuation: (a) V_{power} ; (b) V_{headloss} ; (c) V_{α} ; (d) V_{χ} .

The optimization results obtained by M-I, M-III, and M-IV are presented in Table 4. It is noted that IPOPT fails to find a feasible solution for IEWN II due to the non-linearity of the model, so the results of the M-I serving as feasibility verification are acquired through solving the non-linear water network model with fixed directions given by the solution of M-IV to obtain pump power and then solving the power distribution network NLP model. The results of the objective function gap and violation testify to the scalability of the methods and draw a similar conclusion that M-III is less accurate than M-IV. Though the objective function gap is not noticeable, the quality of the M-III solution is poor, because the violation indices of relaxed quadratic constraints are large. With auxiliary variables introduced by the M-IV method, the variables and constraints increased to 1689 and 4520, resulting in an ascending computational time, which is acceptable. As the testing system grows, relaxation constraints and iteration will inevitably influence the solving time, however, the proposed method is still tractable.

Table 4. Cost results, gaps, and computational time of the three methods under the large-scale IEWN.

Methods	Total cost (\$)	Total cost gap (%)	Computational time (s)
M-I	1833.252	—	—
M-III	1808.539	1.348	2.258
M-IV	1830.528	0.149	6.749

5. Conclusion

This paper proposes a convex reformulation for the non-linear OWPF problem with non-convexity. The accuracy and tightness of the proposed method are evaluated in a small-scale IEWN. The proposed method can solve the OWPF problem in different scenarios, such as heavy loading and load fluctuation. The scalability of the proposed method is also testified on a large-scale IEWN. Simulation results illustrate that the improved MISOCP model and the proposed sequential method outperform the original non-linear formulation and linearized reformulation concerning accuracy, efficiency, and robustness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have included a data availability in the form of a link in the reference list.

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